

**SAFETY DATA SHEET**

Creation Date 16-November-2010

Revision Date 24-December-2021

Revision Number 6

**1. Identification**

**Product Name** Sodium fluoride

**Cat No. :** S299-3; S299-100; S299-500

**CAS-No** 7681-49-4  
**Synonyms** No information available

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

**Details of the supplier of the safety data sheet****Company**

**Importer/Distributor**  
Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

**Manufacturer**  
Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

**Emergency Telephone Number** CHEMTREC®, Inside the USA: 800-424-9300  
CHEMTREC®, Outside the USA: 001-703-527-3887

**2. Hazard(s) identification****Classification**

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                |            |
|------------------------------------------------|------------|
| <b>Acute oral toxicity</b>                     | Category 3 |
| <b>Skin Corrosion/Irritation</b>               | Category 2 |
| <b>Serious Eye Damage/Eye Irritation</b>       | Category 2 |
| <b>Health Hazards Not Otherwise Classified</b> | Category 1 |
| Contact with acids liberates very toxic gas    |            |

**Label Elements**

**Signal Word**  
Danger

**Hazard Statements**  
Toxic if swallowed  
Causes skin irritation  
Causes serious eye irritation

Contact with acids liberates very toxic gas



### Precautionary Statements

#### Prevention

Take any precaution to avoid mixing with acids  
Do not breathe dust/fumes/gas/mist/vapours/spray  
Wear respiratory protection  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Wear protective gloves/protective clothing/eye protection/face protection

#### Response

IF INHALED: Remove person to fresh air and keep comfortable for breathing  
Immediately call a POISON CENTER/doctor  
IF SWALLOWED: Immediately call a POISON CENTER/doctor  
IF ON SKIN: Wash with plenty of soap and water  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Rinse mouth  
Take off contaminated clothing

#### Storage

Store locked up

#### Disposal

Dispose of contents/container to an approved waste disposal plant

## 3. Composition/Information on Ingredients

| Component       | CAS-No    | Weight % |
|-----------------|-----------|----------|
| Sodium fluoride | 7681-49-4 | >95      |

## 4. First-aid measures

|                                        |                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                  | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                              |
| <b>Eye Contact</b>                     | In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.                                                                                                                                                                                                                              |
| <b>Skin Contact</b>                    | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                    |
| <b>Inhalation</b>                      | Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. |
| <b>Ingestion</b>                       | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| <b>Most important symptoms/effects</b> | None reasonably foreseeable. Other side effects may include nausea, headache, diarrhea, cramps, vomiting, or flu-like symptoms                                                                                                                                                                                                 |

## Notes to Physician

Treat symptomatically

## 5. Fire-fighting measures

**Suitable Extinguishing Media** Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

**Unsuitable Extinguishing Media** No information available

**Flash Point** No information available  
**Method -** No information available

**Autoignition Temperature** No information available

**Explosion Limits**

**Upper** No data available

**Lower** No data available

**Sensitivity to Mechanical Impact** No information available

**Sensitivity to Static Discharge** No information available

**Specific Hazards Arising from the Chemical**

Non-combustible. Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

**Hazardous Combustion Products**

Gaseous hydrogen fluoride (HF).

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

**NFPA**

**Health**  
2

**Flammability**  
0

**Instability**  
1

**Physical hazards**  
N/A

## 6. Accidental release measures

**Personal Precautions** Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

**Environmental Precautions** Should not be released into the environment. Do not flush into surface water or sanitary sewer system.

**Methods for Containment and Clean Up** Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

## 7. Handling and storage

**Handling** Wear personal protective equipment/face protection. Avoid dust formation. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe (dust, vapor, mist, gas). Do not ingest. If swallowed then seek immediate medical assistance.

**Storage.** Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from acids. Incompatible Materials. Acids.

## 8. Exposure controls / personal protection

**Exposure Guidelines**

| Component       | Alberta                    | British Columbia           | Ontario TWA/EV             | Quebec                     | ACGIH TLV                  | OSHA PEL                             | NIOSH IDLH                                                |
|-----------------|----------------------------|----------------------------|----------------------------|----------------------------|----------------------------|--------------------------------------|-----------------------------------------------------------|
| Sodium fluoride | TWA: 2.5 mg/m <sup>3</sup> | TWA: 2.5 mg/m <sup>3</sup> | TWA: 2.5 mg/m <sup>3</sup> | TWA: 2.5 mg/m <sup>3</sup> | TWA: 2.5 mg/m <sup>3</sup> | (Vacated) TWA: 2.5 mg/m <sup>3</sup> | IDLH: 250 mg/m <sup>3</sup><br>TWA: 2.5 mg/m <sup>3</sup> |

**Legend**

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH IDLH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Wear safety glasses with side shields (or goggles) Goggles

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Natural rubber | See manufacturers | -               | Splash protection only |
| Nitrile rubber | recommendations   |                 |                        |
| Neoprene       |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

Prevent product from entering drains.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

**9. Physical and chemical properties**

|                                  |                          |
|----------------------------------|--------------------------|
| Physical State                   | Powder Solid             |
| Appearance                       | White                    |
| Odor                             | Odorless                 |
| Odor Threshold                   | No information available |
| pH                               | 7.4                      |
| Melting Point/Range              | 993 °C / 1819.4 °F       |
| Boiling Point/Range              | 1700 °C / 3092 °F        |
| Flash Point                      | No information available |
| Evaporation Rate                 | Not applicable           |
| Flammability (solid,gas)         | No information available |
| Flammability or explosive limits |                          |
| Upper                            | No data available        |
| Lower                            | No data available        |

|                                        |                           |
|----------------------------------------|---------------------------|
| Vapor Pressure                         | 1 mmHg @ 1077 °C          |
| Vapor Density                          | Not applicable            |
| Specific Gravity                       | 2.78 (H <sub>2</sub> O=1) |
| Solubility                             | 40 g/L (25°C)             |
| Partition coefficient; n-octanol/water | No data available         |
| Autoignition Temperature               | No information available  |
| Decomposition Temperature              | No information available  |
| Viscosity                              | Not applicable            |
| Molecular Formula                      | F Na                      |
| Molecular Weight                       | 41.98                     |

## 10. Stability and reactivity

|                                  |                                                        |
|----------------------------------|--------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available             |
| Stability                        | Stable under normal conditions. Moisture sensitive.    |
| Conditions to Avoid              | Incompatible products. Exposure to moist air or water. |
| Incompatible Materials           | Acids                                                  |
| Hazardous Decomposition Products | Gaseous hydrogen fluoride (HF)                         |
| Hazardous Polymerization         | Hazardous polymerization does not occur.               |
| Hazardous Reactions              | None under normal processing.                          |

## 11. Toxicological information

### Acute Toxicity

#### Product Information Component Information

| Component       | LD50 Oral               | LD50 Dermal        | LC50 Inhalation |
|-----------------|-------------------------|--------------------|-----------------|
| Sodium fluoride | LD50 = 52 mg/kg ( Rat ) | > 2000 mg/kg (Rat) | Not listed      |

**Toxicologically Synergistic Products** No information available

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | Irritating to eyes and skin                                                              |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component       | CAS-No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-----------------|-----------|------------|------------|------------|------------|------------|
| Sodium fluoride | 7681-49-4 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known  
**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Other side effects may include nausea, headache, diarrhea, cramps, vomiting, or flu-like symptoms

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Contains a substance which is: Harmful to aquatic organisms. The product contains following substances which are hazardous for the environment.

| Component       | Freshwater Algae                                                                                                  | Freshwater Fish                                                                                                                                                                         | Microtox | Water Flea                                           |
|-----------------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------|
| Sodium fluoride | EC50: = 850 mg/L, 72h static (Desmodesmus subspicatus)<br>EC50: = 272 mg/L, 96h (Pseudokirchneriella subcapitata) | Lepomis macrochirus:<br>530 mg/L LC50 96 h (static)<br>830 mg/L LC50 96 h (semi-static)<br>Pimephales promelas:<br>180 mg/L LC50 96 h<br>Oncorhynchus mykiss:<br>38 - 68 mg/L LC50 96 h | -        | 338 mg/L EC50 = 48 h<br>98 mg/L EC50 = 48 h (static) |

**Persistence and Degradability** Soluble in water Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

UN-No UN1690  
Proper Shipping Name SODIUM FLUORIDE, SOLID  
Hazard Class 6.1  
Packing Group III

### TDG

UN-No UN1690  
Proper Shipping Name SODIUM FLUORIDE, SOLID  
Hazard Class 6.1  
Packing Group III

### IATA

UN-No UN1690  
Proper Shipping Name SODIUM FLUORIDE, SOLID  
Hazard Class 6.1  
Packing Group III

### IMDG/IMO

UN-No UN1690  
Proper Shipping Name SODIUM FLUORIDE, SOLID  
Hazard Class 6.1  
Packing Group III

## 15. Regulatory information

### International Inventories

| Component | CAS-No | DSL | NDSL | TSCA | TSCA Inventory notification - | EINECS | ELINCS | NLP |
|-----------|--------|-----|------|------|-------------------------------|--------|--------|-----|
|-----------|--------|-----|------|------|-------------------------------|--------|--------|-----|

|                 |           |   |   |   | Active-Inactive |           |   |   |
|-----------------|-----------|---|---|---|-----------------|-----------|---|---|
| Sodium fluoride | 7681-49-4 | X | - | X | ACTIVE          | 231-667-8 | - | - |

| Component       | CAS-No    | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-----------------|-----------|-------|----------|------|------|------|------|-------|-------|
| Sodium fluoride | 7681-49-4 | X     | KE-31540 | X    | X    | X    | X    | X     | X     |

**Legend:**

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

**Canada**

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component       | Canada - National Pollutant Release Inventory (NPRI) | Canadian Environmental Protection Agency (CEPA) - List of Toxic Substances | Canada's Chemicals Management Plan (CEPA) |
|-----------------|------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------|
| Sodium fluoride | Part 1, Group A Substance                            |                                                                            |                                           |

**Other International Regulations****Authorisation/Restrictions according to EU REACH**

| Component       | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Sodium fluoride | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component       | CAS-No    | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-----------------|-----------|----------|------------------------------|---------------------------|--------------------------------------------|
| Sodium fluoride | 7681-49-4 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component       | CAS-No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Sodium fluoride | 7681-49-4 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

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**Revision Summary**

This document has been updated to comply with the requirements of WHMIS 2015 to align with the Globally Harmonised System (GHS) for the Classification and Labelling of Chemicals.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**